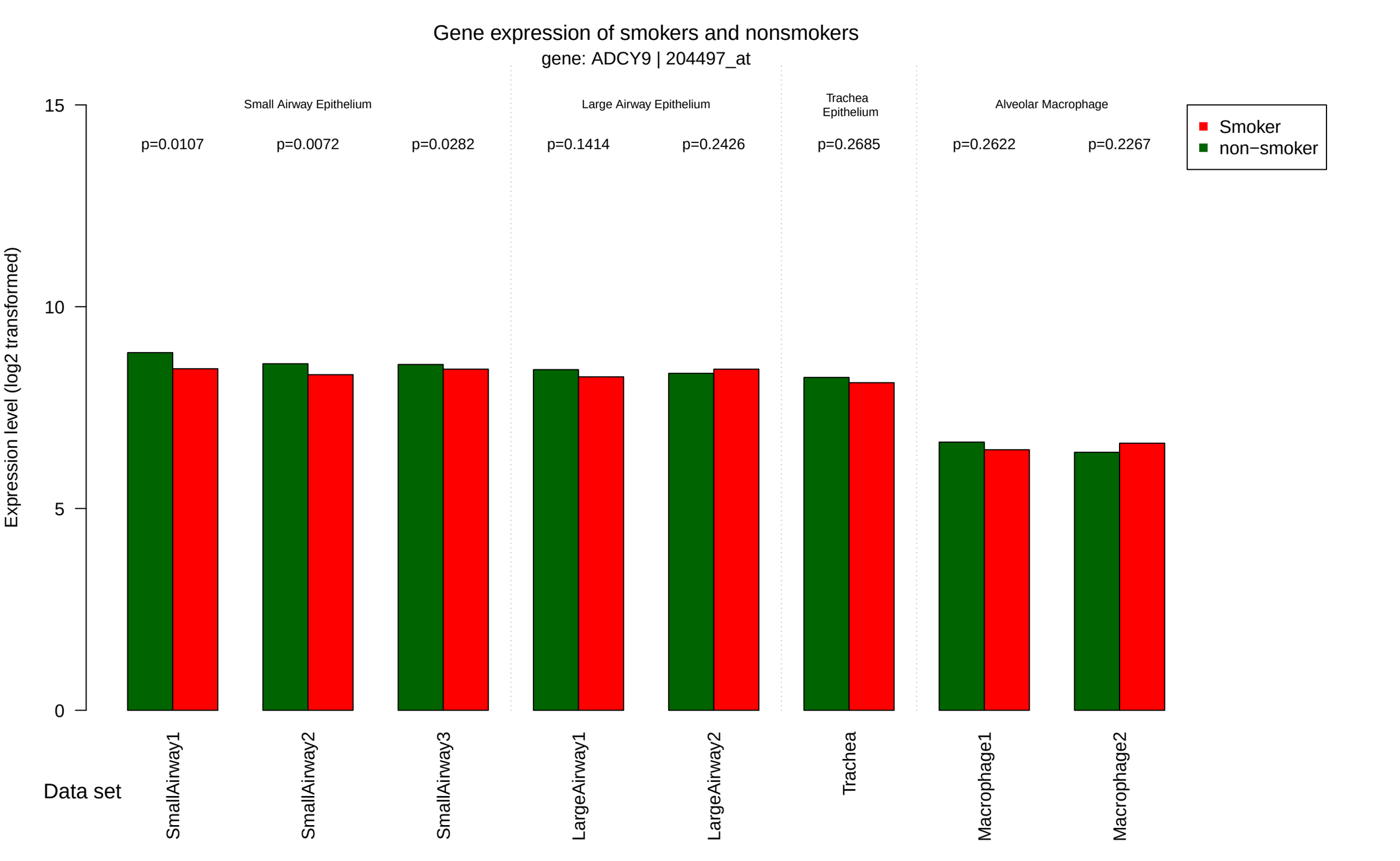
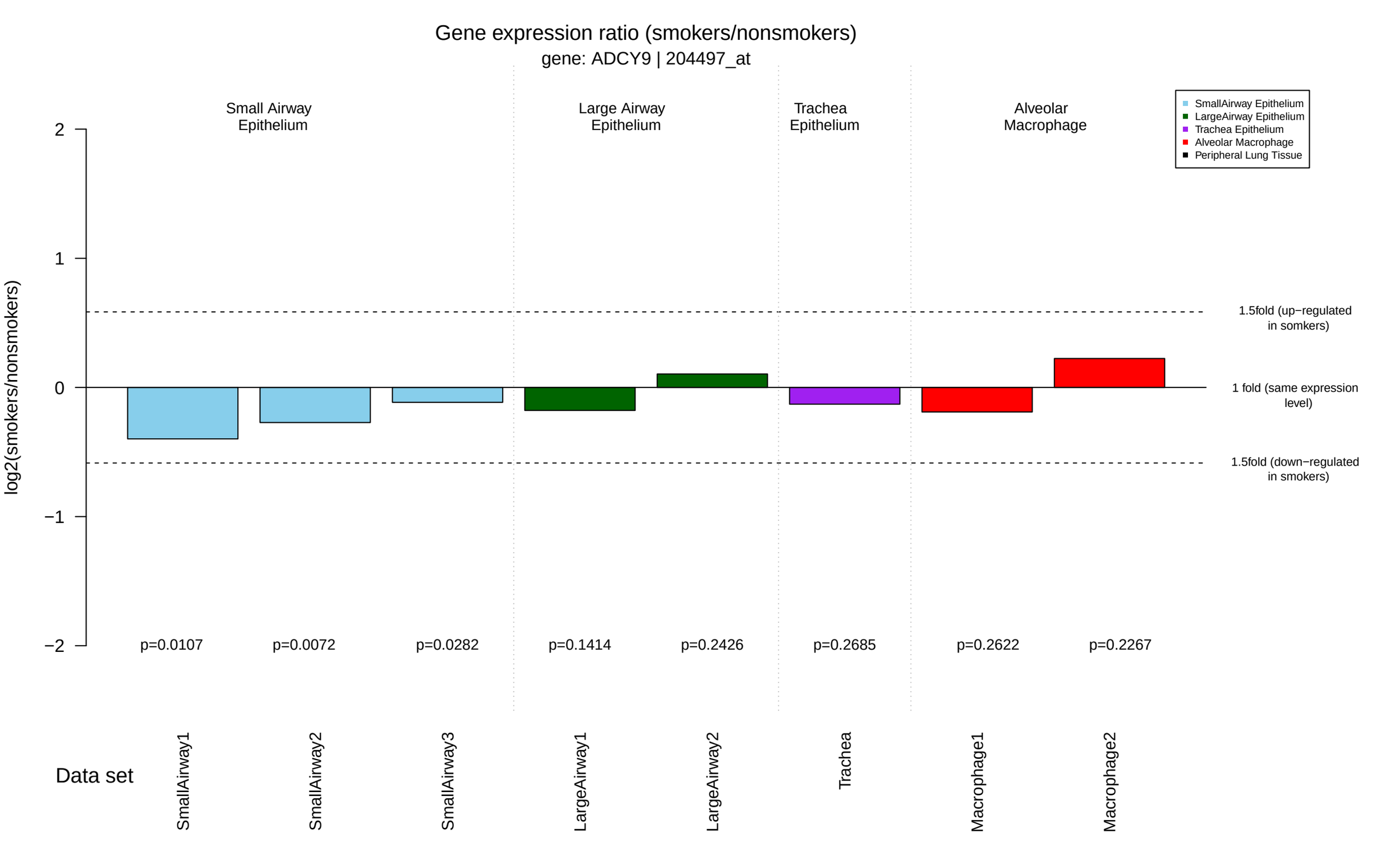
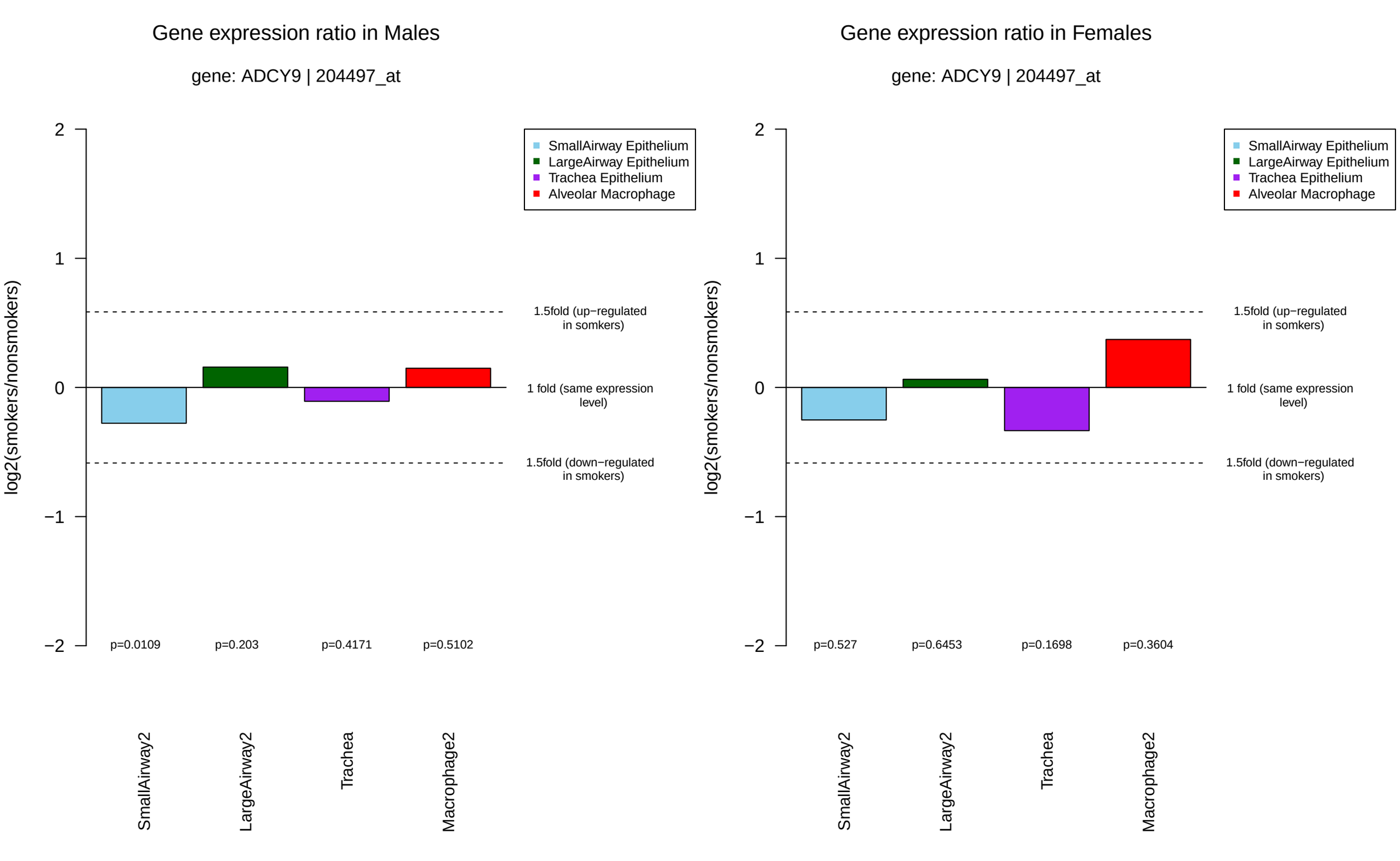
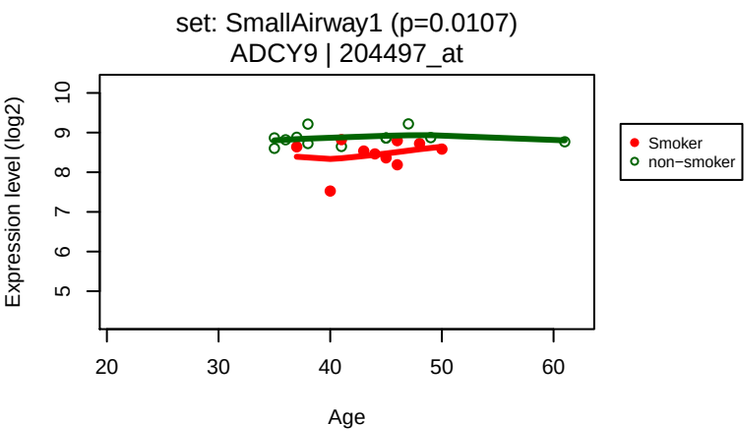
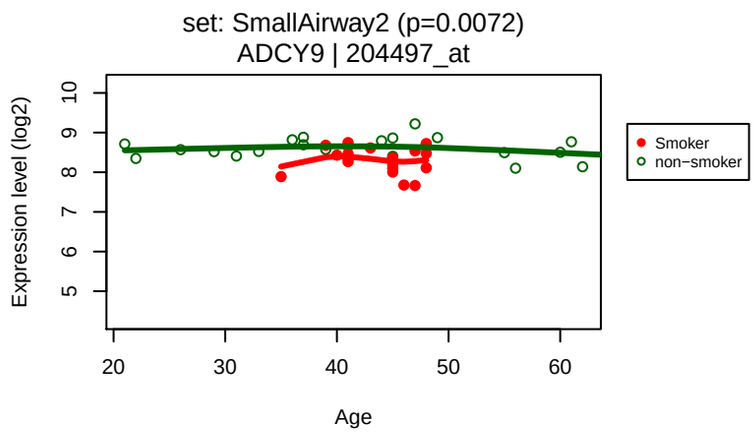
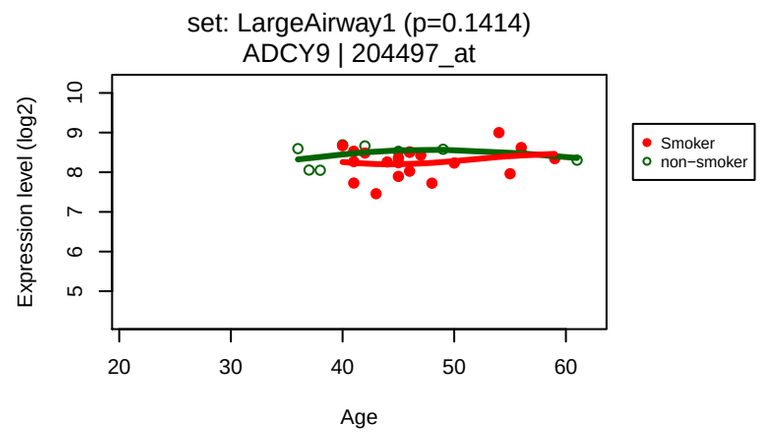

Gene expression of smokers and nonsmokers
gene: ADCY9 | 204497_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0107
p=0.0072
p=0.0282
p=0.1414
p=0.2426
p=0.2685
p=0.2622
p=0.2267
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

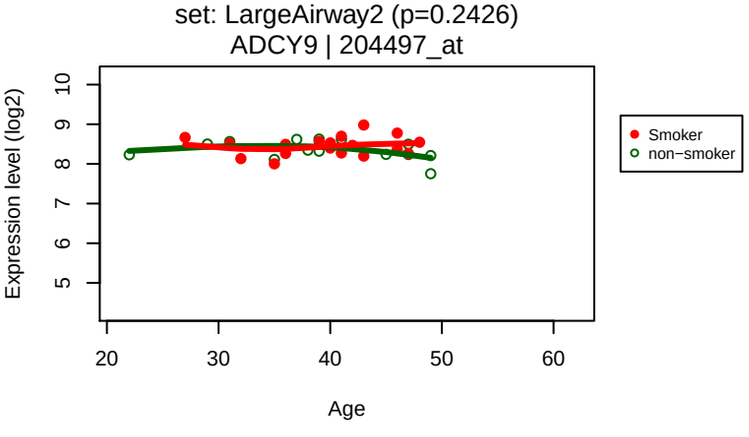

Gene expression ratio (smokers/nonsmokers)
gene: ADCY9 | 204497_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0107
p=0.0072
p=0.0282
p=0.1414
p=0.2426
p=0.2685
p=0.2622
p=0.2267
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

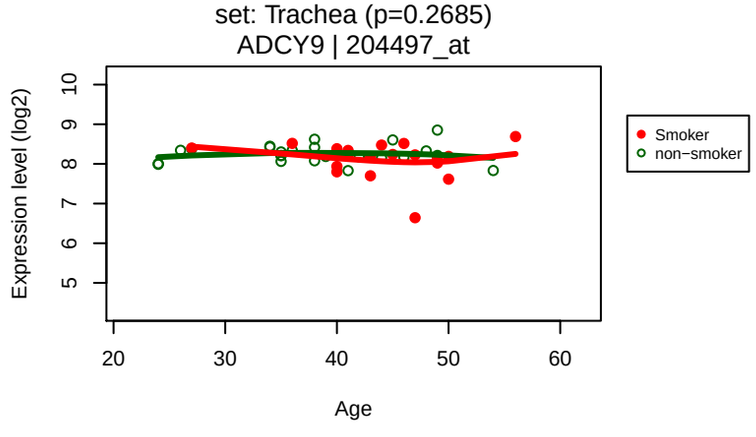

Gene expression ratio in Males
gene: ADCY9 | 204497_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0109
p=0.203
p=0.4171
p=0.5102
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADCY9 | 204497_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.527
p=0.6453
p=0.1698
p=0.3604
SmallAirway2
LargeAirway2
Trachea
Macrophage2

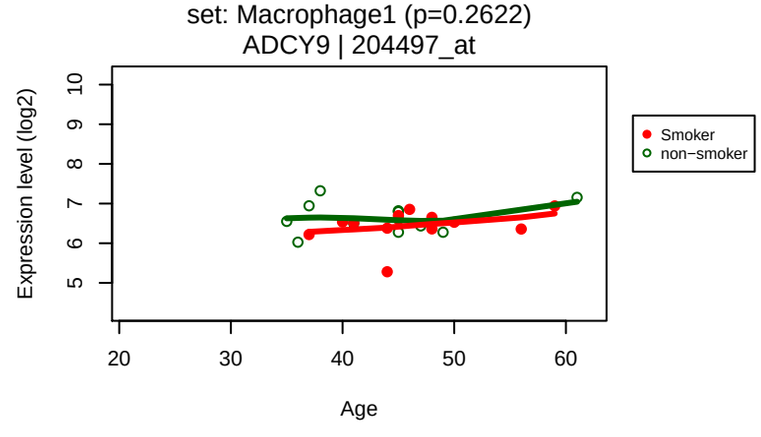

set: SmallAirway1 (p=0.0107)
ADCY9 | 204497_at
Expression level (log2)
Age
Smoker
non−smoker

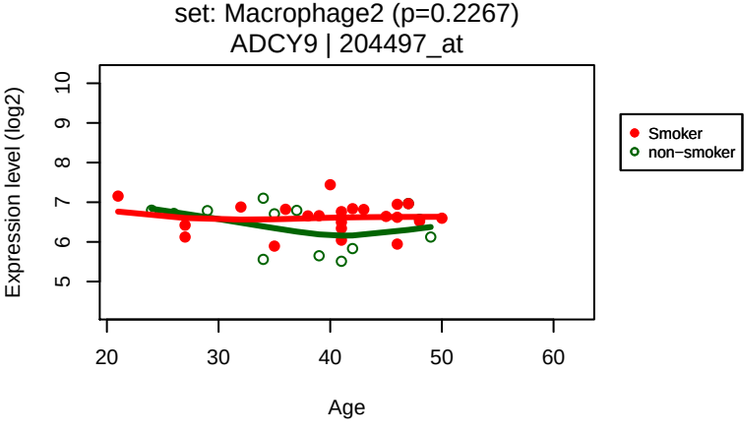

set: SmallAirway2 (p=0.0072)
ADCY9 | 204497_at
Expression level (log2)
Age
Smoker
non−smoker

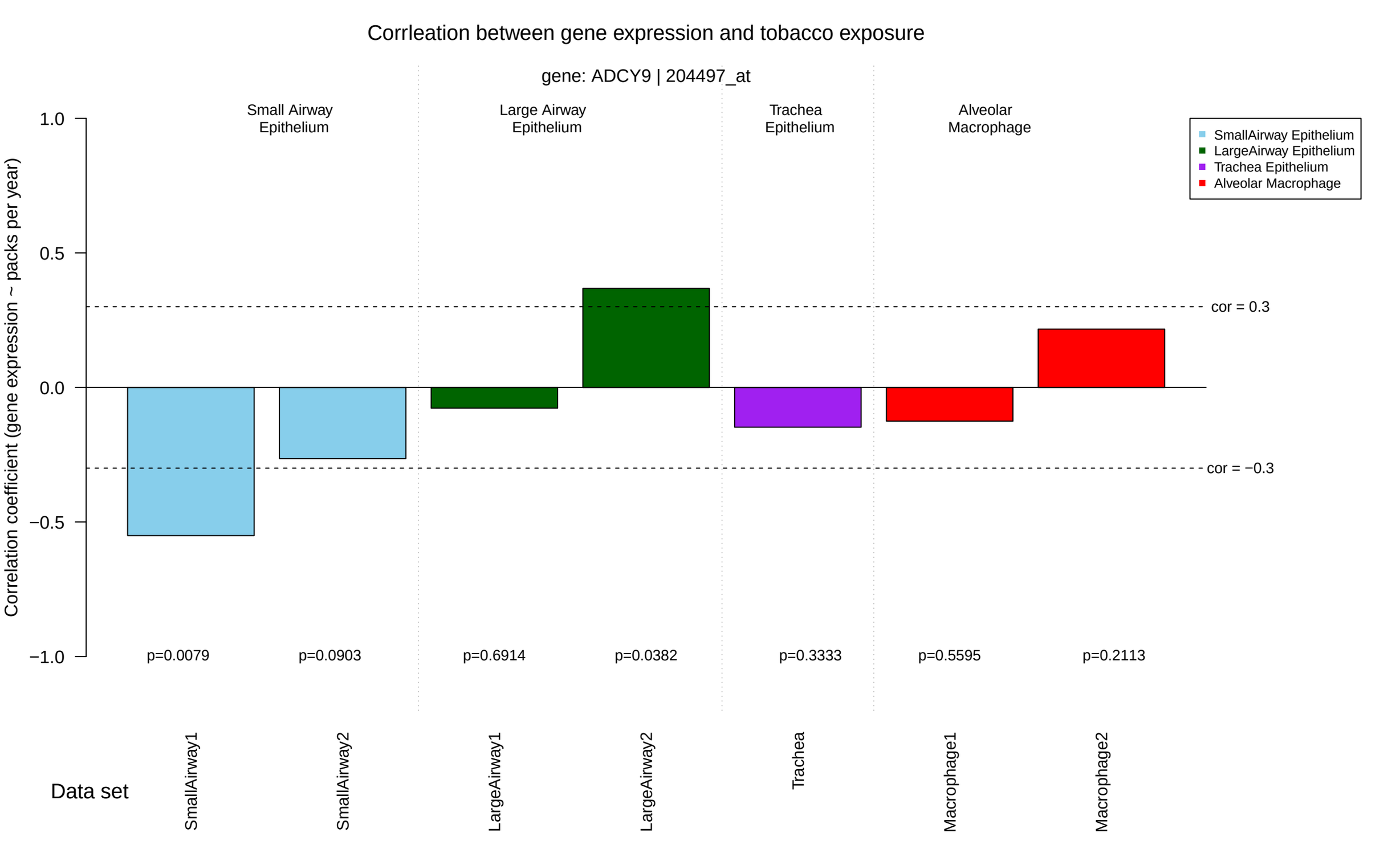

set: LargeAirway1 (p=0.1414)
ADCY9 | 204497_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2426)
ADCY9 | 204497_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2685)
ADCY9 | 204497_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2622)
ADCY9 | 204497_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2267)
ADCY9 | 204497_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADCY9 | 204497_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0079
p=0.0903
p=0.6914
p=0.0382
p=0.3333
p=0.5595
p=0.2113
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2